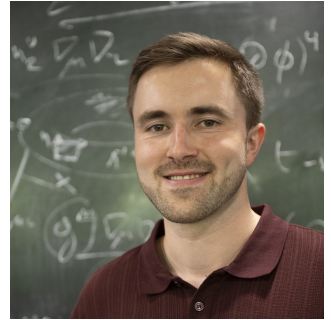


SEPPE J. STAELENS



CONTACT INFO

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LinkedIn www.linkedin.com/in/seppestaelens

ABOUT ME

I am a motivated, young researcher with a profound interest in theoretical physics and astrophysics. I am especially excited about topics that combine both fields, like gravitational waves and black holes. In my research, I leverage multiple computational techniques varying from numerical relativity to ray tracing. Furthermore, I enjoy teaching mathematics and physics to both high school and University students.

EDUCATION

PhD in Applied Mathematics and Theoretical Physics **2023-2027**
University of Cambridge, DAMTP | Cambridge, United Kingdom

- Numerical relativity, boson stars, black holes and gravitational waves
- Funding: Centre for Doctoral Training in Data Intensive Science (courses on research computing, Bayesian statistics and machine learning)

MSc Astronomy & Astrophysics **2020-2023**
KU Leuven | Leuven, Belgium

- Graduated *summa cum laude* (87.00 %)
- Thesis: "Merging Compact Objects in the LISA Frequency Band" under supervision of P. Jonker, G. Nelemans. Grade: 17.8/20
- Erasmus exchange to Radboud Universiteit Nijmegen (2022-2023)

MSc Physics **2020-2022**
KU Leuven | Leuven, Belgium

- Graduated *summa cum laude with congratulations of the Board of Examiners* (90.63 %)
- Thesis: "Black Hole Photon Rings beyond General Relativity" under supervision of T. Hertog, D. Mayerson, F. Bacchini. Grade: 18.1/20

TWIN BA Physics & Mathematics **2017-2020**
KU Leuven | Leuven, Belgium

- Combination of two BA degrees for excelling students
- Graduated (phys.) with *magna cum laude* (84.42 %)
- Graduated (math.) with *summa cum laude* (85.17 %)

EXPERIENCE

DATA SCIENCE INTERN (PART-TIME) <i>FactTrace AI Cambridge, UK</i>	2026
• Development of fingerprinting infrastructure for claim verification • Deep learning, AI, high-dimensional data analysis.	
DIRAC INNOVATION PLACEMENT <i>Cancer Research UK Scotland Institute Glasgow, UK</i>	2025
• Project: <i>Multimodal data integration and analysis</i> • Machine / Deep Learning in a GPU-accelerated HPC environment • Cancer biology, bioinformatics, high-dimensional data analysis	

PUBLICATIONS

In case of equal contributions, authors are listed alphabetically.

Book Chapter	[B1] Lupsasca, A., D. R. Mayerson, B. Ripperda, and S. Staelens (2024). "A Beginner's Guide to Black Hole Imaging and Associated Tests of General Relativity". In: <i>Recent Progress on Gravity Tests. Challenges and Future Perspectives</i> . Ed. by C. Bambi and A. Cardenas-Avendano, pp. 183-237. doi: 10.1007/978-981-97-2871-8_6 . arXiv: 2402.01290 [gr-qc] .
Publications	[P7] Staelens, S. J. and E. Alkan (June 2026). "Ray tracing ultra-compact boson stars: visibility modulations and incomplete photon rings". Submitted to PRD. arXiv: 2606.30729 [gr-qc]. [P6] Marks, G. A., S. J. Staelens, and U. Sperhake (Apr. 2026). "Black Hole-Boson Star Binaries: Gravitational Wave Signals and Tidal Disruption". Submitted to PRD. arXiv: 2604.06312 [gr-qc]. [P5] Canizares, P., S. J. Staelens, and I. Romero-Shaw (Feb. 2026). "The Shape of Eccentricity: Rapid Classification of Eccentric Binaries with the Wavelet Scattering Transform". Submitted to PRD. arXiv: 2602.24079 [gr-qc]. [P4] Staelens, S. J. (Feb. 2026). "Search for growing angular modes in ultracompact boson star evolutions". In: <i>J. Phys. Conf. Ser.</i> 3177.1, p. 012046. doi: 10.1088/1742-6596/3177/1/012046. arXiv: 2508.19460 [gr-qc]. [P3] Marks, G. A., S. J. Staelens, T. Evstafyeva, and U. Sperhake (2025). "Long-Term Stable Nonlinear Evolutions of Ultra-compact Black-Hole Mimickers". In: Phys. Rev. Lett. 135.13, p. 131402. doi: 10.1103/lk48-7r2f. arXiv: 2504.17775 [gr-qc]. [P2] Staelens, S. and G. Nelemans (2024). "Likelihood of white dwarf binaries to dominate the astrophysical gravitational wave background in the mHz band". In: <i>Astron. Astrophys.</i> 683, A139. doi: 10.1051/0004-6361/202348429. arXiv: 2310.19448 [astro-ph.HE]. [P1] Staelens, S., D. R. Mayerson, F. Bacchini, B. Ripperda, and L. Küchler (2023). "Black hole photon rings beyond general relativity". In: <i>Phys. Rev. D</i> 107.12, p. 124026. doi: 10.1103/PhysRevD.107.124026. arXiv: 2303.02111 [gr-qc].
Master's Theses	[MT2] Staelens, S. (2023). "Merging Compact Objects in the LISA Frequency Band". Link to PDF . MA thesis. KU Leuven.

[MT1] **Staelens, S.** (2022). "Black Hole Photon Rings beyond General Relativity". [Link to PDF](#). MA thesis. KU Leuven.

TALKS

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|--------------------------|---|
| Seminar | <ul style="list-style-type: none">• <i>Long-term stable non-linear evolutions of ultracompact black hole mimickers</i>. Perimeter Institute for Theoretical Physics, Waterloo (CA), May 08, 2025. Recording.• <i>Stable evolutions of ultracompact black hole mimickers</i>. Utrecht University, Utrecht (NL), Oct 30, 2025. |
| Invited Talks | <ul style="list-style-type: none">• <i>The shape of eccentricity: rapid classification of eccentric binaries with the wavelet scattering transform</i>. CamGW Meeting, Cambridge (UK), Jun 4, 2026. Recording. |
| Recorded Talks | <ul style="list-style-type: none">• <i>Stable evolutions of ultracompact BH mimickers in 3+1 dimensions</i>. 24th International Conference on General Relativity and Gravitation & the 16th Edoardo Amaldi Conference on Gravitational Waves, Glasgow (UK), July 15, 2025. Recording |
| Contributed talks | <ul style="list-style-type: none">• 2026, Cancer Research UK: Scotland Institute, Glasgow, UK• 2025, Massachusetts Institute of Technology, Cambridge MA, USA• 2025, Belgian-Dutch GW meeting, Radboud University Nijmegen, NL• 2025, Canadian Institute for Theoretical Astrophysics, Toronto, Canada• 2025, Numerical Relativity and Fundamental Fields, Trieste, Italy• For a full list of talks, see this webpage. |

TEACHING AND MENTORING

STUDENT MENTORING 2025-2026

University of Cambridge | Cambridge, UK

- Supervised an undergraduate student during a joint research project.
- Project resulted in a joint paper [P7].

SUPERVISOR / TEACHING ASSISTANT 2024-2026

University of Cambridge | Cambridge, UK

- *General Relativity - Part II of the Math. Tripos*
- *Numerical Relativity and Gravitational Waves - Part III of the Math. Tripos*

ACADEMIC TUTOR 2024,2025

Immerse Education | Cambridge, UK

- 2-week long summer school in mathematics
- Designed own 40-hour course

Other teaching	• <i>Physics at University</i> (4 lectures). St. Andrew's College Cambridge (2025) • Summer School: Oxbridge Academic Bootcamp (2025) • Teaching assistant. Cambridge Centre for International Research (2023). • Teaching Assistant. Faculty of Sciences, KU Leuven (2020-2022). • Study Support Services. Faculty of Science, KU Leuven (2020, 2022).
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OUTREACH

Events	Six public engagement events (2023-2026), including Pint of Science, Cambridge Festival, and primary-school outreach (NRICH, CHaOS).
Podcasts	• 2026, Mar 11. Voices of Mathematics. Episode 5, <i>Black holes, dark stars and gravitational waves: Ulrich Sperhake and Seppe Staelens</i>. Listen on podbean. • 2025, Sep 14. Einstein Telescope: ET voor de vrienden (Dutch). Episode 2: <i>A black hole on the highway</i>. More info on the website.
Articles	• Co-authored two Dutch popular science articles on the white-dwarf gravitational-wave background (Zenit, NTvN, 2025). • Press releases: NOVA (2025), Faculty of Mathematics (2026) and Plus maths (2026).

ACHIEVEMENTS

Awards	• Prize for best Master's Thesis in Physics 2022 (400 EUR) • 9th place in the <i>Flemish Geography Olympiad 2017</i>
Scholarships	• Erasmus+ scholarship: study abroad for one year. • Fully funded PhD scholarship (4y.). Centre for Doctoral Training in Data Intensive Science, funded through STFC. (~ £200.000).
Certificates	Youth Leader (2016), Head Youth Leader (2018), and Instructor (2021). Issued by the Flemish Government.

EXTRACURRICULARS

CUBS	Cambridge University Belgian Society. President 2024-2026, Secretary 2023-2024.
Sporta	An organization for youth camps, of which I have been a part for 7 years. I have been youth leader at 20+ summer camps, which taught me teamwork and responsibility. Furthermore, I was an instructor at 5 youth leader courses, which taught me how to transfer knowledge in multiple ways.

SKILLS

● ● ●	GRChombo, FOORT, Python, $\LaTeX$
● ● ○	C++, Linux, HPC, PyTorch
● ○ ○	Mathematica, FORTRAN

Languages Dutch (native), English (C2), French (B2)

REFERENCES

Ulrich Sperhake	Professor Department of Applied Mathematics and Theoretical Physics Centre for Mathematical Sciences University of Cambridge 01223 766861 us248@damtp.cam.ac.uk
Gijs Nelemans	Professor Department of Astrophysics Radboud University Nijmegen +31 24 365 2983 nelemans@astro.ru.nl
Daniel Mayerson	Postdoctoral researcher Institute for Theoretical Physics Department of Physics & Astronomy KU Leuven +32 16 37 68 90 daniel.mayerson@kuleuven.be
Thomas Hertog	Professor Institute for Theoretical Physics Department of Physics & Astronomy KU Leuven +32 16 32 72 46 thomas.hertog@kuleuven.be